



Lakshya JEE 2.0 (2024)

Probability

DPP-01

1. Four numbers are chosen at random from $\{1, 2, 3, \dots, 40\}$. The probability that they are not consecutive, is
 - (1) $\frac{1}{2470}$
 - (2) $\frac{4}{7969}$
 - (3) $\frac{2469}{2470}$
 - (4) $\frac{7965}{7969}$
2. Probability of product of a perfect square when 2 dice are thrown together, is
 - (1) $\frac{2}{9}$
 - (2) $\frac{1}{9}$
 - (3) $\frac{5}{18}$
 - (4) None of these
3. Probability of all 3-digit numbers having all the digits same is
 - (1) $\frac{1}{100}$
 - (2) $\frac{3}{100}$
 - (3) $\frac{7}{100}$
 - (4) None of these
4. Probability of throwing 16 in one throw with three dice is
 - (1) $\frac{1}{36}$
 - (2) $\frac{1}{18}$
 - (3) $\frac{1}{72}$
 - (4) $\frac{1}{9}$
5. There are 4 addressed envelopes and 4 letters. Then the chance that all the letters are not mailed through proper envelope is:
 - (1) $\frac{1}{24}$
 - (2) 1
 - (3) $\frac{23}{24}$
 - (4) $\frac{9}{2}$
6. Three numbers are chosen at random without replacement from $\{1, 2, 3, \dots, 10\}$. The probability that the minimum of the chosen number is 3 or their maximum is 7, is:
 - (1) $\frac{11}{26}$
 - (2) $\frac{7}{20}$
 - (3) $\frac{11}{40}$
 - (4) $\frac{7}{40}$
7. A determinant of second order is made with the elements 0 and 1. What is the probability that the determinant is positive?
 - (1) $\frac{7}{12}$
 - (2) $\frac{11}{12}$
 - (3) $\frac{3}{16}$
 - (4) $\frac{15}{16}$
8. A number k is selected from the set $\{1, 2, 3, 4, \dots, 10\}$. If $k^2 - 4k + 3 < 0$ then the probability is
 - (1) $\frac{1}{5}$
 - (2) $\frac{2}{5}$
 - (3) $\frac{3}{10}$
 - (4) $\frac{1}{10}$
9. There are 2 vans each having numbered seats, 3 in the front and 4 at the back. There are 3 girls and 9 boys to be seated in the vans. The probability of 3 girls sitting together in a back row on adjacent seats, is:
 - (1) $\frac{1}{13}$
 - (2) $\frac{1}{39}$
 - (3) $\frac{1}{65}$
 - (4) $\frac{1}{91}$
10. The first 12 letters of English alphabet are written down at random in a row. The probability that there are exactly 4 letters between A and B, is:
 - (1) $\frac{7}{33}$
 - (2) $\frac{7}{66}$
 - (3) $\frac{7}{99}$
 - (4) $\frac{5}{33}$

11. Six boys and six girls sit in a row randomly. The probability that the six girls sit together or the boys and girls sit alternately, is
- (1) $\frac{3}{308}$ (2) $\frac{1}{100}$
 (3) $\frac{2}{205}$ (4) $\frac{4}{407}$
12. If A , B and C are mutually exclusive and exhaustive events associated with a random experiment.
 If $P(B) = \frac{3}{2}P(A)$ and $P(C) = \frac{1}{2}P(B)$, then $P(A)$ is equal to
- (1) $\frac{2}{13}$ (2) $\frac{4}{13}$
 (3) $\frac{6}{13}$ (4) $\frac{8}{13}$
13. If A and B are two events such that
 $P(A \cap B) = \frac{1}{4}$, $P(\bar{A} \cap \bar{B}) = \frac{1}{5}$ and $P(A) = P(B) = p$,
 then p is equal to:
- (1) $\frac{17}{40}$ (2) $\frac{19}{40}$
 (3) $\frac{21}{40}$ (4) $\frac{23}{40}$
14. If two events A and B are such that
 $P(\bar{A}) = 0.3$, $P(B) = 0.4$ and $P(A \cap \bar{B}) = 0.5$, then
 $P\left(\frac{B}{A \cup \bar{B}}\right)$ is equal to
- (1) $\frac{1}{4}$ (2) $\frac{1}{5}$
 (3) $\frac{2}{5}$ (4) $\frac{3}{5}$
15. Two dice are thrown. The probability that the number appeared have a sum of 8, if it is known that the second die always exhibits 4, is
- (1) $\frac{5}{6}$ (2) $\frac{1}{6}$
 (3) $\frac{2}{3}$ (4) $\frac{1}{3}$
16. If $P(A) = \frac{3}{8}$, $P(B) = \frac{5}{8}$ and $P(A \cup B) = \frac{3}{4}$, then
 $P\left(\frac{\bar{A}}{\bar{B}}\right)$ is equal to
- (1) $\frac{1}{4}$ (2) $\frac{1}{3}$
 (3) $\frac{2}{3}$ (4) $\frac{3}{4}$
17. An urn contains 8 red and 5 white balls. Three balls are drawn at random. Then the probability that balls of both colours are drawn is
- (1) $\frac{40}{143}$ (2) $\frac{70}{143}$
 (3) $\frac{3}{13}$ (4) $\frac{10}{13}$
18. Four fair dice D_1 , D_2 , D_3 and D_4 each having six faces numbered 1, 2, 3, 4, 5 and 6 are rolled 'simultaneously'. The probability that D_4 shows a number appearing on one of D_1 , D_2 and D_3 , is
- (1) $\frac{91}{216}$ (2) $\frac{108}{216}$
 (3) $\frac{125}{216}$ (4) $\frac{127}{216}$
19. Words with or without meaning are to be formed using all the letters of the word EXAMINATION. The probability that the letter M appears at the fourth position in any such word is:
- (1) $\frac{1}{66}$
 (2) $\frac{1}{11}$
 (3) $\frac{1}{9}$
 (4) $\frac{2}{11}$
20. A speaks the truth in 60% cases and B in 70% cases. The probability that they will say the same thing while describing a single event is:
- (1) 0.56 (2) 0.54
 (3) 0.38 (4) 0.94

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (3)
2. (1)
3. (1)
4. (1)
5. (3)
6. (3)
7. (3)
8. (4)
9. (4)
10. (2)

11. (1)
12. (2)
13. (3)
14. (1)
15. (2)
16. (3)
17. (4)
18. (1)
19. (2)
20. (2)



PW Web/App - <https://smart.link/7wwosivoicgd4>

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